Journal of Systems Thinking

Systems Engineering-Reengineered

The Fourth Wave

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> *"I will be frank. Educators, and I include myself, for I have spent many years as an adjunct professor at various institutions, are far less certain how to teach "generalship" than we are of how to teach the laws of thermodynamics. And yet it is clear that an understanding of the broad issues, the big picture, is so much more influential in determining the ultimate success or failure of an enterprise than is the mastery of any given technical detail. The understanding of the organizational and technical interactions in our systems, emphatically including the human beings who are a part of them, is the present-day frontier of both engineering education and practice."*
>
> ***Michael Griffin, Boeing Lecture, Purdue University, 28 March 2007***

The word *system* comes from the Greek words *sun* and *histanal* meaning '*with*' and '*set-up*' respectively. Combining these two words became *sustema*, which subsequently evolved into the French word *system.* At first glance, the Greek roots of *system* may seem peculiar, but *set-up with* (or *with set-up*) captures the essence of a principal challenge when designing systems – *integration*. Specifically, as a system is "a set of different elements so connected or related as to perform a unique function not performable by the elements alone," ensuring internal components are set-up to interact with each other and their external environment is essential [1]. As the world becomes more complex and connected, the ability to integrate systems will become even more essential and difficult. Systems Engineering was born out of this necessity and challenge in the early- and mid-20th century, as complicated aerospace and defense systems demanded deliberate architecting and integration.

Systems Engineering is a field that concerns itself with realizing successful systems. It was born of a need to develop complicated aerospace and defense systems in the early-and mid-20th century and is increasingly essential in today's interconnected world. To realize successful systems, engineers use Systems Thinking, itself a descendant of the field, to "[understand the] broad issues, the big picture… to [determine] the ultimate success or failure of an enterprise" [2]. This is seen very clearly in the successful development of the iPhone. Steve Jobs wanted to impact a customer's experience with a device. The development of that device, which emphasized controlling all aspects of the system design, was a complex Systems Engineering endeavor. From physical development to software development, there were thousands of mini-processes that had to be integrated and synchronized in order for the iPhone to be introduced.

Systems Engineering, an interdisciplinary field sitting at the nexus of system sciences and engineering, focuses on designing and managing complex systems over their life cycles. At its core, it utilizes Systems Thinking principles to help its practitioners "[understand the] broad issues, the big picture … [and to determine] the ultimate success or failure of an enterprise" [2]. The speed of innovation continues to increase. As a result, the value of Systems Engineering and Systems Engineers will increase with the increasing complexity of modern systems.

With this in mind, this chapter provides the historical origins of Systems Engineering, discusses how it is currently being taught, codifies some necessary core competencies, and illustrates the power of the discipline through a case study.

Systems Engineering

The Need for and Growth of Systems Engineering

> *In simpler systems, the methods and point of view of Systems Engineering have little to offer more effective than those that have long been known and used by traditional engineering specialties. It is when the systems become complex that the methods of Systems Engineering will be most profitable, and where the traditional engineering disciplines have little to offer [, thus] the emphasis in Systems Engineering on* ***extremely complex systems*** *[3].*

Our world is growing more complex and this complexity is driven by the increasing interconnectedness among natural, social, and technological systems [4,5]. Though perhaps individually simple, the connections among these constituents drives a complexity that must be addressed—either harnessed or mitigated—to achieve desirable results. While this phenomenon is widely recognized today across a variety of domains (e.g., environmental with global warming, social with economic bubbles and bursts, etc.), it was first realized in the realm of technology and addressed with the development of Systems Engineering.

A "technology" (i.e., a human designed system) can be seen as the result of the combination of other technologies [6]. This evolutionary, combinatorial nature of technologies means they will become increasingly nested and more complicated over time. For centuries, and even millennia, technological growth has been limited and slow [7]. This allowed people—individuals, organizations, societies—to adapt organically to technological change, and the implications of that change. Furthermore, the nature of these combinations, and therefore the complexity of technologies, was limited. This is partly a function of the limited number of technologies, and partly a function of the limited size of the organizations (and populations) developing new technologies.

All of this changed in the beginning of the 20th century, which saw a sharp increase in the amount and complexity of technological development. The increasing complexity of the technological systems, the size of the organizations developing these systems, and the speed at which they were developed made organic technological development untenable. Consider anecdotally, the evolution of human flight. From the dawn of civilization until December 16, 1903, mankind only dreamed of flying under its own power.

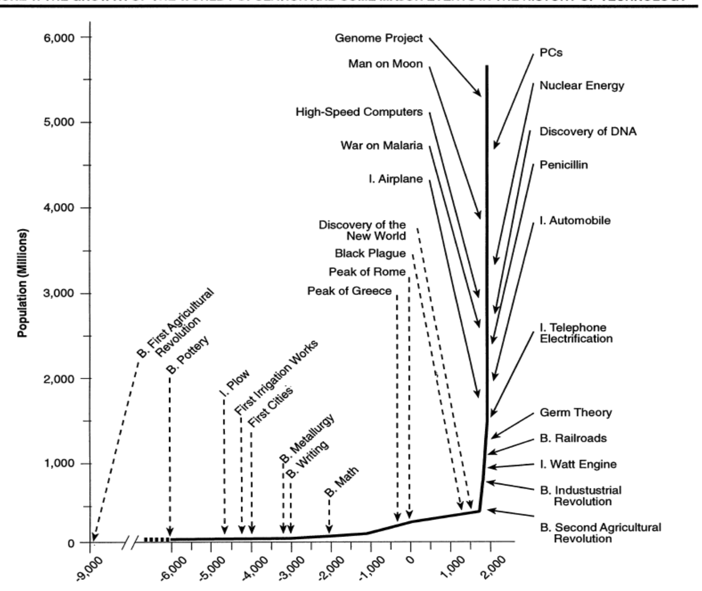
On December 17, 1903, Orville and Wilbur Wright realized a successful heavier-than-air flying system—the airplane. They were certainly not the first to attempt to realize such a system; there is an extensive history of attempted flight documented by the American Institute of Aeronautic and Astronautics (AIAA). The Wright brothers were merely the first successful team. With less than 50 contributors and clear decision authority, they learned organically through trial and error, adopting what worked and discarding what did not. The technology, while revolutionary, was not terribly complex, and through their tinkering a new era took flight.

Contrast this with another of mankind's achievements only 63 years later - walking on the moon. In several decades as opposed to millennia, mankind made a quantum leap, but this time the scale of the system - the Apollo missions - was enormous. NASA estimates that approximately 400,000 people were necessary to realize the Apollo system [8]. While the Apollo missions are exceptional, this trend toward increasingly complex systems being developed at an increasing rate by increasingly large teams has held. To contend with this challenge, the field of Systems Engineering emerged.

Systems Engineering is a young but growing field. Unlike other engineering fields, which harness base scientific knowledge to realize human needs (e.g., electrical engineering, which applies electricity and magnetism, or chemical engineering, which applies chemistry), Systems Engineering was born of a need to manage and harness social, environmental, and technological complexity [3].

Accordingly, Systems Engineering did not exist in any real way prior to the 20th century, or arguably until the 1930s; however, it has grown exponentially since its inception. Figure 2 approximates the academic growth of both Systems Engineering and Systems Thinking by the annual number of publications in the Web of Science's Core Collection with each term appearing in the topic field. As the growth of publications in general has increased over the decades, the grey line in the figure indicates the hypothetical growth that should have been seen for these two disciplines over the latter half of the 20th century, if they had simply grown with the population of Web of Science publications.

FIGURE 1. THE GROWT:I OF THE WORLD POPULATION AND SOME MAJOR EVENTS IN THE HISTORY OF TECHNOLOGY



Sources: Cipolla (1974), Clark (1971), Fagan (1977), McNeill (1971), Piggott (1965), and Trewartha (1969). See also Allen (1992, 1994), Slicher van Bath (1963), and Wrigley (1987).

Notes: I = invention; B = beginning. There is usually a lag between the invention of a process or a machine and its general application to production. "B" is intended to identify the beginning, or the earliest stage, of this diffusion process.

Figure 1: Number of Publications in Clarivate's Web of Science Core Collection with Systems Engineering or Systems Thinking in the Topic Field.

Systems Engineering Definition(s) and Scope

> *Systems Engineering is an interdisciplinary approach and means to enable the realization of successful systems. It focuses on defining customer needs and required functionality early in the development cycle, documenting requirements, then proceeding with design synthesis and system validation while considering the complete problem: Operations, Performance, Test, Manufacturing, Cost & Schedule, Training & Support, Disposal.*
>
> *Systems Engineering integrates all the disciplines and specialty groups into a team effort forming a structured development process that proceeds from concept to production to operation. Systems Engineering considers both the business and the technical needs of all customers with the goal of providing a quality product that meets the user needs. [9]*

Blaise Pascal famously lamented that had he had more time, he would have written a shorter letter; E.F. Schumacher noted that genius is necessary for simplicity [10]. These ideas express a universal truth—that clarity and simplicity are preferable to verbosity and complexity, yet harder to achieve and time-consuming to develop. As a young field, therefore, it is understandable that the definition(s) of Systems Engineering are verbose, complex, and not universally accepted. A short, simple, and widely accepted definition will surely arise, but it will take time and a touch of genius.

As seen in Table 1, there are many proposed definitions of Systems Engineering, and each has its own nuance. Nonetheless, the definition that began this section generally captures their collective essence, and its significant points are summarized below:

- *Value-Based*: Ultimately, Systems Engineering is about developing systems that are useful to people and organizations.
- *Requirements-Driven*: To realize a successful system, one must define success in a way that can be engineered (i.e., requirements).
- *Interdisciplinary and Integrative*: Systems Engineering is inherently interdisciplinary and has a focus on integrating the results of multiple disciplines.
- *Immersive*: Systems Engineering must consider the system in the context of its environment.
- *Longitudinal*: Systems Engineering must look across the life-cycle of a system - how it performs not just in operation, but also in development, production, maintenance, and retirement.
- *Holistic and Managed*: Systems Engineering and systems engineers must lead and manage other disciplines to balance competing requirements. That is, local optimization does not generally yield global optimization or even global realization of a system.

Finally, it is important to note that while there are many fields denoted as “X Systems Engineering,” where X may be some particular field (e.g., Computer Systems Engineering, Control Systems Engineering, or Biological Systems Engineering), none of the definitions in the table are definitions for “X Systems Engineering.” They are for “Systems Engineering” writ large, as it is domain agnostic when it is implemented in some domain (e.g., defense, health care, communications, etc.). Put another way, Systems Engineering has a home in any field with complex systems, where the need to define the problem, organize and integrate teams and disciplines, and assess global performance of the system in a systemic and systematic manner is paramount.

<table>
<tr><th>Year</th><th>Definition</th><th>Citation</th></tr>
<tr><td>1955</td><td>“A systems engineer is a man who, when assigned to a project, is capable of coordinating and preparing or carrying out three things - first, an optimum scheme (system survey) showing the interrelation of all factors that will affect the project and sway future action; second, an optimum set of organized ideas (the plan of procedure) pertaining to carrying out the project; and third, an optimum physical realization (the product) of the devices necessary to terminate the project.”</td><td>[11]</td></tr>
<tr><td>1993</td><td>Systems engineers, in contrast [to other engineering disciplines], are problem staters.
The principal functions of Systems Engineering are:
to develop statements of system problems comprehensively…,
to resolve top-level system problems into simpler problems that are solvable by technology: hardware, software, and bioware, and
to integrate the solutions to the simpler problems into systems to solve the top-level problem.</td><td>[12]</td></tr>
<tr><td>2006</td><td>Engineering of a System.
Engineering discipline that develops, matches, and trades off requirements, functions, and alternate system resources to achieve a cost-effective, life-cycle-balanced product based upon the needs of the stakeholders</td><td>[13]</td></tr>
</table>

Table 1. Definitions of Systems Engineering from the Field.

History of Systems Engineering

The term *Systems Engineering* is generally considered to have been coined by the Bell Telephone laboratories in the 1940s [14], and its history is well documented (e.g., [9,14–19]. The field grew rapidly, particularly in the aerospace and defense fields, and perhaps its crowning achievement of the field was placing a man on the moon through the Apollo program. It was formalized in the defense sector in 1969 with the establishment of MIL-STD-499 and has been used widely across the aerospace and defense sectors ever since.

Arguably, two defining aspects of a field are the establishment of educational and research programs in universities and the establishment of professional societies. In academia, Systems Engineering was first formally taught in the 1950's and 1960's. Notably, MIT developed a course of instruction in 1950[15], and the University of Arizona established the first Systems Engineering department in 1961[20]. As of 2017, INCOSE reports that 33 U.S. universities and colleges offer undergraduate programs in Systems Engineering, and 60 have masters and doctorate programs [21]. These numbers are generally confirmed by a query of the National Center for Education Statistics' Integrated Postsecondary Education Data System (IPEDS), which indicates 26 U.S. schools granted bachelor's degrees in Systems Engineering in 2017, while 75 schools conferred master's degrees. In either case, the conclusion is clear - Systems Engineering, as an academic discipline, is on the rise, and this is further confirmed in Figures 3 and Figure 4, which show the annual number of Systems Engineering degree-granting schools and degrees conferred between 1987 and 2017. Of note, in Figure 3 the dashed lines represent the number of Systems Engineering degrees that would have been conferred if the conferment grew in accordance with the annual growth rate in degrees by type. As the theoretical growth is substantially below the realized growth, Systems Engineering appears in vogue. Moreover, the number of Accreditation Board for Engineering and Technology (ABET)-accredited Systems Engineering programs has increased nearly three-fold over the past decade, attesting to the quality of the programs.

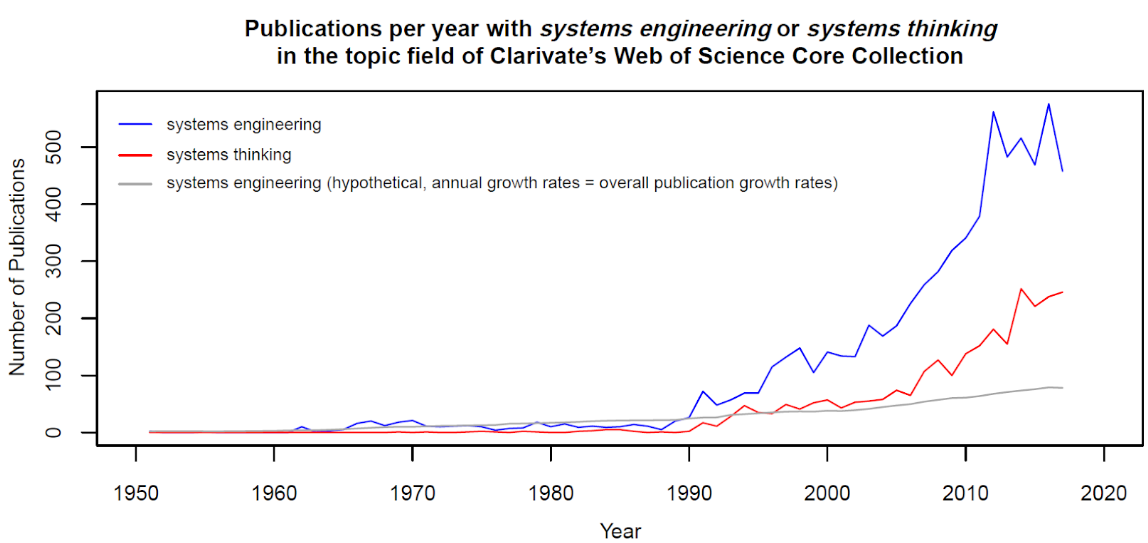


Figure 2: Number of Schools Granting Systems Engineering Degrees between 1987 and 2017.

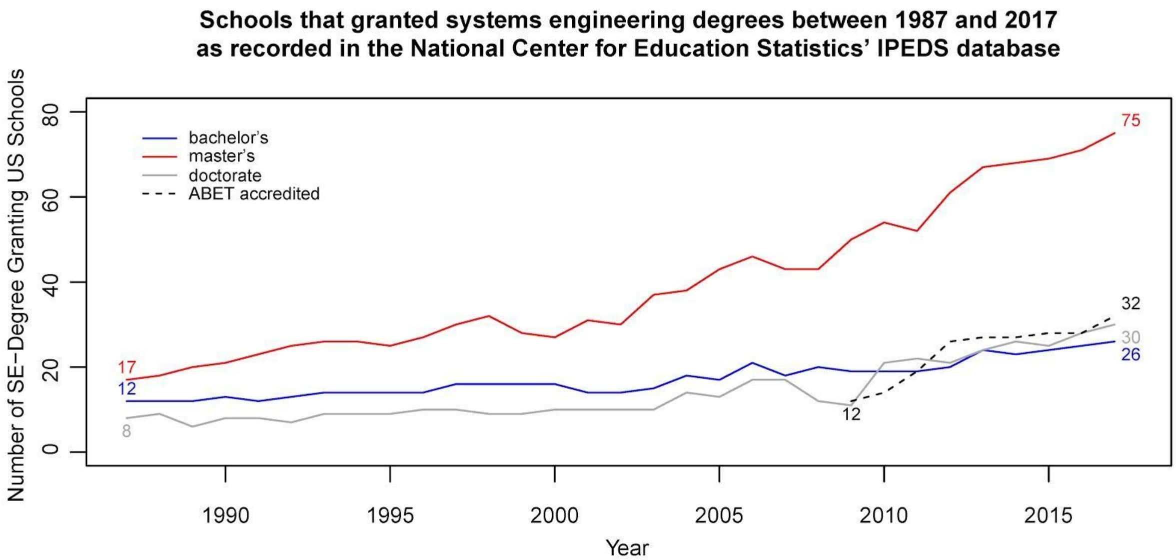


Figure 3: Number of Systems engineering Degrees Conferred between 1987 and 2017.

With respect to professional societies, ABET currently recognizes seven organizations as co-leads for Systems Engineering, namely: the American Society of Mechanical Engineers (ASME), the Computing Sciences Accreditation Board (CSAB), the Institute of Electrical and Electronics Engineers (IEEE), the Institute of Industrial and Systems Engineers (IISE), the International Society of Automation (ISA), the Society of Automotive Engineers (SAE) International, and INCOSE. As this diverse list attests, Systems Engineering is inherently interdisciplinary. Among these societies, INCOSE is arguably the most relevant and inclusive. With over 17,000 members, an annual conference, and a peer-reviewed journal devoted to the discipline, INCOSE is subspecialty-agnostic, and it has added structure and formalism to the field by helping to maintain a Systems Engineering Body of Knowledge (SEBoK) and certifying systems engineering professionals [22].

The Core Systems Engineering Competencies

Systems Engineering sits in the "integrative-middle" of the engineering disciplines. To become an expert in Systems Engineering, there are two sets of core competencies that need to be cultivated over time; Engineering Competencies and Systems Thinking Competencies (depicted in Figure 4). We build on the work done by Body of Knowledge and Curriculum to Advance Systems Engineering (BKCASE) and the Systems Engineering Research Center's Helix Project. Both highlight the critical knowledge, skills and attributes needed to be effective systems engineers, with a key attribute being the ability to translate multiple engineering disciplines into consumable and accessible information that can enable solving complex problems (SERC). This requires a depth of understanding of multiple engineering disciplines and breadth of knowledge of tools that can be leveraged across engineering disciplines. This presents a T-model of system engineering due to the development of breadth and depth in Systems Engineering theories, models, and tools [23,24]. These can be referred to collectively as the "Engineering Competencies".

The Engineering Competencies include numerous theories, methods and tools in modern engineering. The "T" model of Systems Engineering develops the breadth and depth in Systems

Engineering theories, methods, and tools while simultaneously developing sufficient depth in the mechanical, civil, and electrical engineering disciplines, for example as described in Table 2.

Civil Engineering	The oldest engineering focused on building things (construction)
Electrical Engineering	Developed in the 19th century to deal with technology and electrical innovation (circuits)
Mechanical Engineering	Versatile form engineering focused on objects and systems in motions (automotive engines)
Systems Engineering	Interdisciplinary form of engineering in focused on design and management of complex systems (ecosystems)

Table 2: Differences in Engineering Discipline.

The specific theories, methods, and tools are ever-evolving, but the systems engineer combines depth and breadth in integrative systems design with sufficient depth in the engineering disciplines, making them ideally suited to lead. To be successful, systems engineers must be:

1) Systematic - follow some organized, repeatable methodology.
2) Systemic - understand the whole of a system.

The complexity of the world requires a systems engineer to be able to approach ambiguity to clarify a way forward. Leveraging Systems Thinking is essential because each situation is distinctly different and a systems engineer must be able to transfer prior experience to the new unknown. As discussed earlier, there are many types of Systems Thinking. We highlight the Distinction, Systems, Relationship, and Perspective (DSRP) framework as a way to approach Systems Thinking. These are broadly "Systems Thinking Competencies" [4,25,26]. We are informed by prior work that describes systems thinking skills and competencies, and focus on the DSRP model due to its flexibility in its approach to provide structure to ambiguity. The Systems Thinking Competencies include a pattern of thought that provides the basis for a systematic understanding of real-world systems and problems. Systems Thinking is an iterative process that seeks better approximation of real-world phenomena by coupling mental models with constant real-world feedback. These competencies are therefore basal to all problem solving, design and engineering systems. Figure 4 provides a visual description of the T-Shaped concept applied to Engineering Competencies and how one can digest the Systems Thinking Competencies.

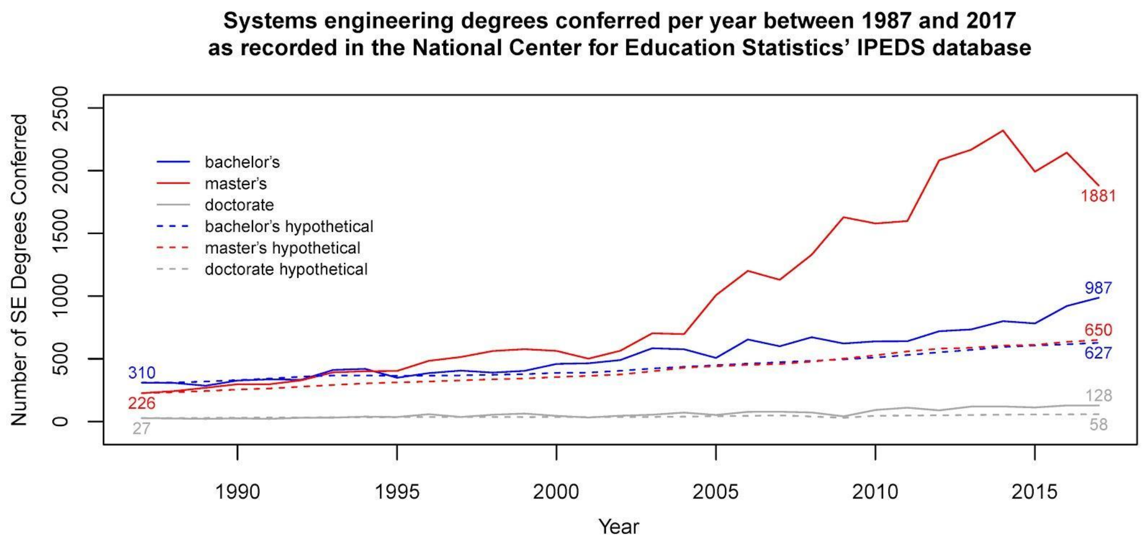


Figure 4: Engineering Competencies and Systems Thinking Competencies.

When we combine the Engineering and Systems Thinking competencies, the "Core Systems Engineering Competencies" are defined. The Core Competencies of Systems Engineering sit in the "integrative middle" of the engineering disciplines, theories, practices, and tools. The systems engineer leverages Systems Thinking to understand the system, define the problem, iterate to real-world feedback, and discover the solution.

Because there are many possible entry points to the problem or system, where to start is not always clear in Systems Engineering. The entry point is determined by the problem. Systems engineers traverse spheres ("wormholes") in order to discover solutions. These three basic spheres—technical, social, and political—could be broken down further into sub-spheres. However, solving any problem begins with understanding the system. Core Competencies are applied at the beginning and throughout. Figure 5 illustrates how this might be observed in systems engineers.

Where do we go from here?

Beyond addressing current issues, INCOSE has also outlined a vision for the future of Systems Engineering, which stresses, among other things, the need to engineer versus evolve the *system of systems* (SoS) implied by the emerging Internet of Things (IoT) [27]. According to multiple, recent studies (e.g., [28,29]), the number of connected devices is expected to explode in the coming decade, and the interfaces between and interaction of these devices will fall under the purview of system engineers. Moreover, as the interdependencies between systems grow, it will become increasingly challenging to draw an analytical boundary around SoS. In this environment, the need for Systems Thinking becomes paramount, and INCOSE's *Systems Engineering Vision 2025* [27], as well as the recent, dramatic growth of Systems Thinking publications, (as seen in Figure 1), support this assertion. Interestingly, an excellent example of engineering a highly connected device is the development of Apple's iPhone, which addressed the interface and boundary challenge by effectively isolating the system inside a closed architecture. While this approach tends to counter the current move towards modular designs, it highlights the value of deliberate integration within a well-defined solution space.

Case Study

In 2007, Apple launched the iPhone. This device arguably represented the most significant personal communications breakthrough in history. Within years, smartphones proliferated the globe and today there are nearly 3 billion devices used *worldwide*. Steve Jobs' and Apple's vision upon launching the device was extraordinarily bold: change the world and change the communications paradigm of the global citizen. That is exactly what happened. Apple's achievements, and the impact of the iPhone, are now in the record books of telecommunications and computing history. The methods that enabled this accomplishment, and the principles that underpinned their achievements, are not as well understood. In part, this is due to the strict secrecy that shrouds much of Apple's most important work. However, there is one aspect of Apple's product development history that is apparent through examination and analysis, and this one aspect is probably the most important component of the iPhone's success: *exceptional Systems Engineering*.

Supporting this assertion requires some discussion of "exceptional systems engineering." One of the more significant problems involving Systems Engineering involves ambiguity and often-outright disagreement across the profession regarding the definition of Systems Engineering [16]. However, themes emerge that the profession tends to echo and embrace. Blanchard and Fabrycky [16] describe several of these themes:

- a "top down approach that views the system as a whole";
- a "life-cycle orientation that addresses all phases to include system design and development, production and/or construction, distribution, maintenance and support, retirement, phase-out, and disposal";
- "a better and more complete effort regarding initial definition of system requirements"; and
- "an interdisciplinary team approach throughout the system design and development process."

Exceptional Systems Engineering at Apple, particularly for the iPhone, centered on three principles: (1) design first, engineer second; (2) insist on holistic design; and (3) leadership matters in Systems Engineering. These three principles alone do not encapsulate a sufficient Systems Engineering methodology. However, these three principles do represent the core of what makes Apple different, and successful. Apple's emphatic pursuit of these three principles allowed many other key principles of systems engineering to fall into place.

Design first, engineer second. Apple has always been infatuated with design. Jony Ive, the well-known "designer-in-chief" at Apple, holds more power than many of the most senior executives at Apple. Jobs once famously stated, "The only person at Apple with more operational power than me is Jony Ive" [30]. This "design-first" emphasis at Apple manifests across numerous qualities and functions commonly experienced with Apple devices: multi-touch screen technology; swiping and scrolling; brushed aluminum exterior; sleek, aesthetic physical and graphics appearance; intuitive and consistent software; satisfying haptic feedback; and seamless integration across devices. The essence of Systems Engineering involves the realization of a system that meets stakeholder needs. If Systems Engineering remains moored in stakeholder needs, user-expectation failures and disappointments become much less likely. Constraints, feasibility analysis, and known solutions can overtake the system when engineering drives design. This can create solutions that disappoint stakeholders. When design drives engineering, stakeholder needs remain at the forefront and drive the engineering to a desirable solution. Ive states that this type of design philosophy requires product development to occur in a more “collaborative, integrated way.” This collaboration and integration amongst engineers lead to a system solution that emerges from a strong, concurrent, interdisciplinary design.

Insist on holistic design. Holistic design at Apple started with Jobs’ experiences with the company in the 1980s. Jobs was a known perfectionist and obsessed with control. This often created a troublesome managerial style; however, it also led to a masterfully integrated system. Computer engineers constantly try to merge disparate software and hardware and create systems capable of adopting and adapting to outside technology. Jobs was completely averse to this idea. Jobs believed in absolute control of the software and the hardware. In his mind, this philosophy was essential to optimize system performance [31]. The design philosophy of the Mac, iPod, iPhone, and iPad all reflect this desire to closely control the entirety of the system – holistic system design. This insistence on holistic design, which ultimately included the supply chain, manufacturing, packaging, and distributing, created integration quality that competitors could not achieve simply because their extent of product control was not nearly as complete. Whenever Apple dabbled in a venture that relied extensively on the integration of another company, particularly from a hardware or software standpoint, the resulting product never met the high expectations of Jobs [31]. Isaacson states that “Jobs believed in the ‘coherence’ thing. His faith in a closed and controlled environment remained unwavering, even as Android gained market share.” Jobs’ insistence on holistic design resulted more from his philosophical ideals than his business goals. He was passionate about the entire user experience, and he felt the only way to optimize that experience was through complete control of the entire system.

Leadership matters in Systems Engineering. Steve Jobs may be the most iconic individual to emerge from the 21st century technology boom. While it is inarguable that the innovative success of Apple emerged from the technological genius of many, the product success of Apple relied extensively on the leadership of Steve Jobs. Jobs was notoriously demanding, a perfectionist, and brutally honest [31]. He insisted on the right design driving the solution. His philosophy of holistic design, largely resulting from his compulsive controlling tendencies, resulted in seamless integration. And his technical depth and willingness to learn and evolve enabled him to both anticipate and lead ongoing engineering.

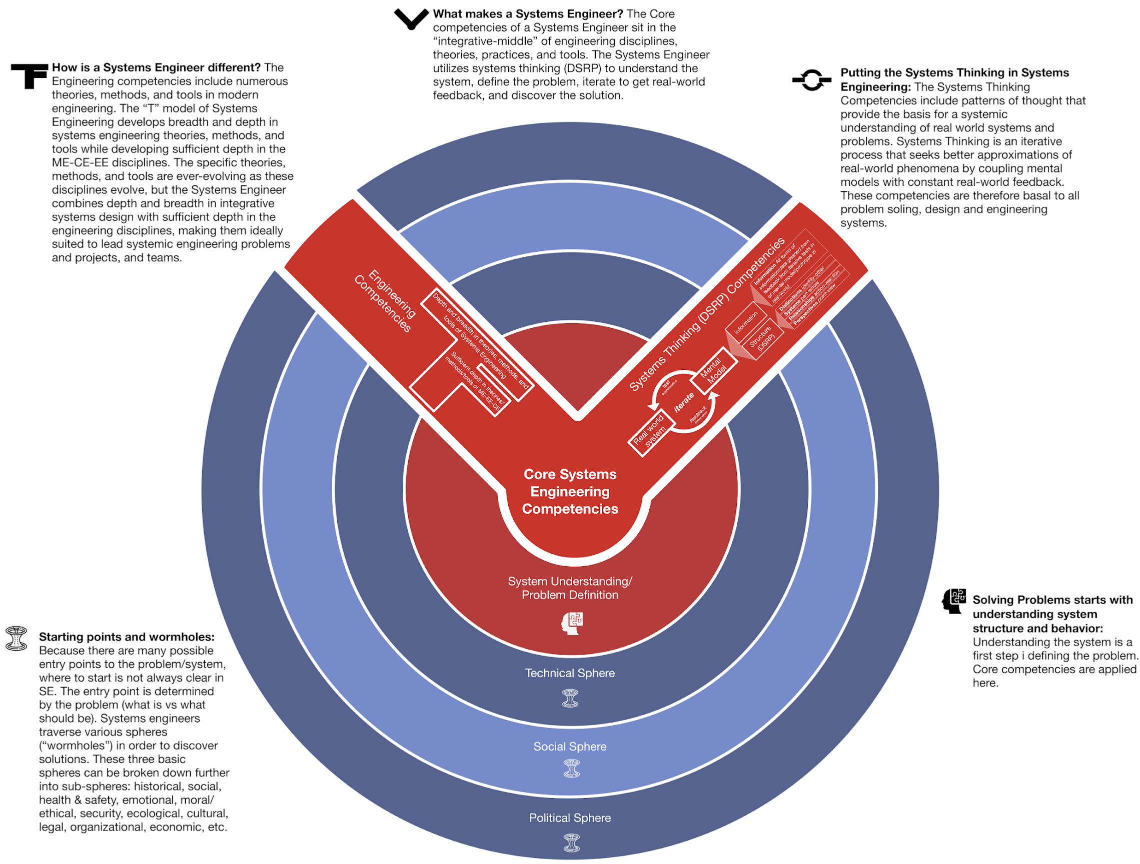


Figure 5: The Core Systems Engineering Competencies.

Conclusion

Systems Engineering was born of the need to engineer complicated technical systems. The methods and tools to do this had applicability beyond the technical world and, in part, spun off as a related, but distinct field of Systems Thinking that was mainly applied to social systems. With the advent of the internet, the social and technical worlds have become inextricably linked. Engineers must now not only concern themselves with the technical, but the social as well as demonstrated by Jobs and the iPhone. Systems Engineering is an ever-evolving field, but it will continue to be of great importance to addressing societies wants and needs. Table 3 revisits the previous definitions and provides an additional definition derived from the model presented. ***Systems Engineering can be defined as that which systems engineers do which is derived from the SE competencies.***

Year	Definition	Citation
1955	"A systems engineer is a man who, when assigned to a project, is capable of coordinating and preparing or carrying out three	[11]

	things - first, an optimum scheme (system survey) showing the interrelation of all factors that will affect the project and sway future action; second, an optimum set of organized ideas (the plan of procedure) pertaining to carrying out the project; and third, an optimum physical realization (the product) of the devices necessary to terminate the project."	
1993	Systems engineers, in contrast [to other engineering disciplines], are *problem staters.* The principal functions of Systems Engineering are: ● to develop statements of system problems comprehensively…, ● to resolve top-level system problems into simpler problems that are solvable by technology: hardware, software, and bioware, and ● to integrate the solutions to the simpler problems into systems to solve the top-level problem.	[12]
2018	IBM does not provide a precise definition, but has this: "Systems Engineering - Implement compliant, traceable solutions that manage a wide range of requirements to achieve faster deployment. ● Requirements Management ● Agile Software Development ● Model-based Systems Engineering ● Test management ● Functional safety and compliance"	[32]
2019	**Systems Engineering can be defined as that which systems engineers do which is derived from the SE competencies.**	

Table 3. Definitions of Systems Engineering from the Field.

The next chapter will build on the leadership component of the core competencies presented in this chapter to provide a holistic perspective of the core engineering competencies.

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